(Please write your Enrollment Number) Enrollment No. 0140102022

End-Term Examination (Regular & Reappear)
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech ECE>, Semester:<3>
(November-December, 2023)

Subject Code: BEC 203	Subject: Signals & Systems

Time :3 Hours **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8=20)	
	(a)	Difference between Static and Dynamic.		
	(b)	What do you mean by Superposition principle ? Explain it with example.		
	(c)	Check whether the signals are periodic or aperiodic-: I. Sin 30 πt II. Sinπt u(t)		
	(d)	Explain the property Time shifting with example.		
	(e)	Explain the difference between Linear and Non-Linear phase.		
	(f)	Describe the properties of LTI system.		
	(g)	Explain the properties of ROC.		
	(h)	What do you mean by Aliasing?		
		UNIT-I		
Q2	State the Differences with example-: a) Even and Odd signals b) Continuous time signal and discrete time signals		(10)	
Q3	Draw the waveforms of the following signals-: a) $X_1(t)=u(t-2)$ b) $X_2(t)=u(t+3)$ c) $X_3(t)=u(-2t+1)$ d) $X_4(t)=u(-t)$ e) $X_5(t)=u(t)-2u(t-1)+u(t-3)$		(10)	
		UNIT-II		
Q4	Describe the properties of periodic signals in detail		(10)	
Q5	Find the Fourier transform of the signal g(t) defined by $g(t)=te^{-at}u(t)$		(10)	
		UNIT-III		
Q6	Explain the terms-: a) Group Delay b) Phase Delay		(10)	
Q7	What do you mean by All Pass system? Explain it with suitable diagrams.		(10)	
		UNIT-IV		
Q8	Explain the terms-: a) Sampling b) Interpolation		(10)	
Q9	Determine the Laplace transform of-: $X(t)= -e^{-2t}u(-t)+e^{-3t}u(-t)$		(10)	